

Experiment 13: Host Discovery

Commands in Kali Linux

Aim

To execute Nmap host discovery commands in Kali Linux to identify active hosts on a network.

Procedure

1. Open the Kali Linux terminal.
2. Check the Nmap installation using `nmap --version`.
3. Enter each host discovery command separately.
4. Use an authorized lab network, such as `192.168.1.0/24`.
5. Observe and record the output.

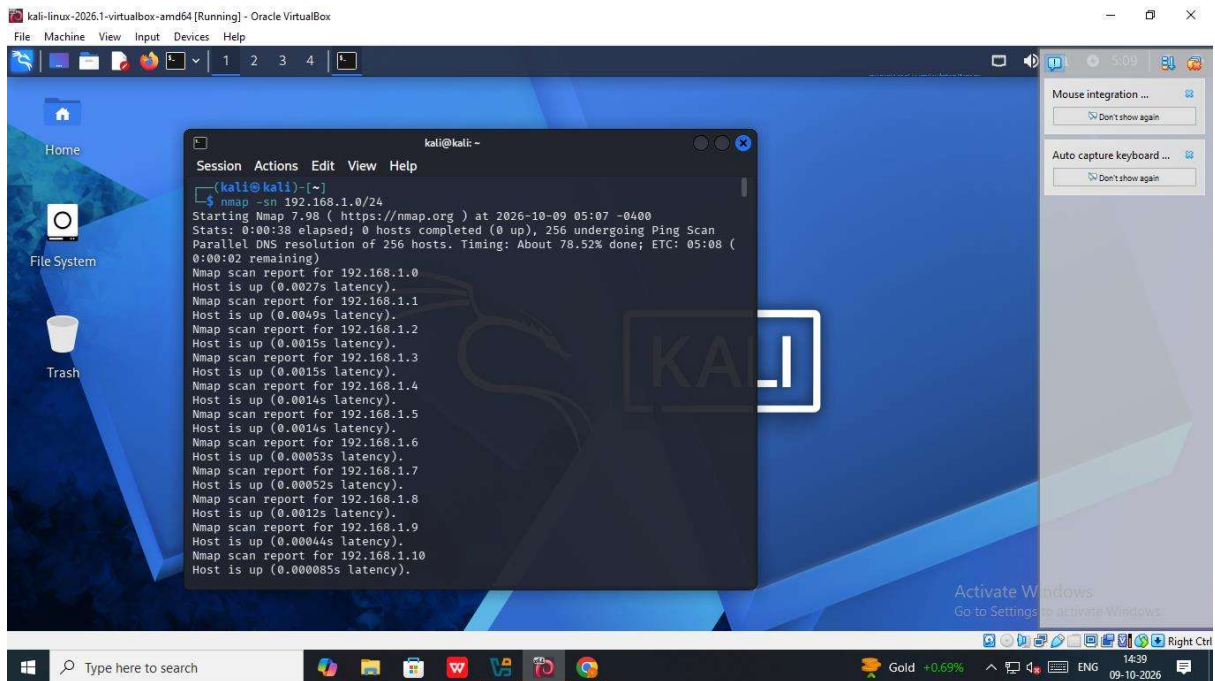
Input and Output

Input 1: Ping Scan

Command:

```
nmap -sn 192.168.1.0/24
```

Output 1:

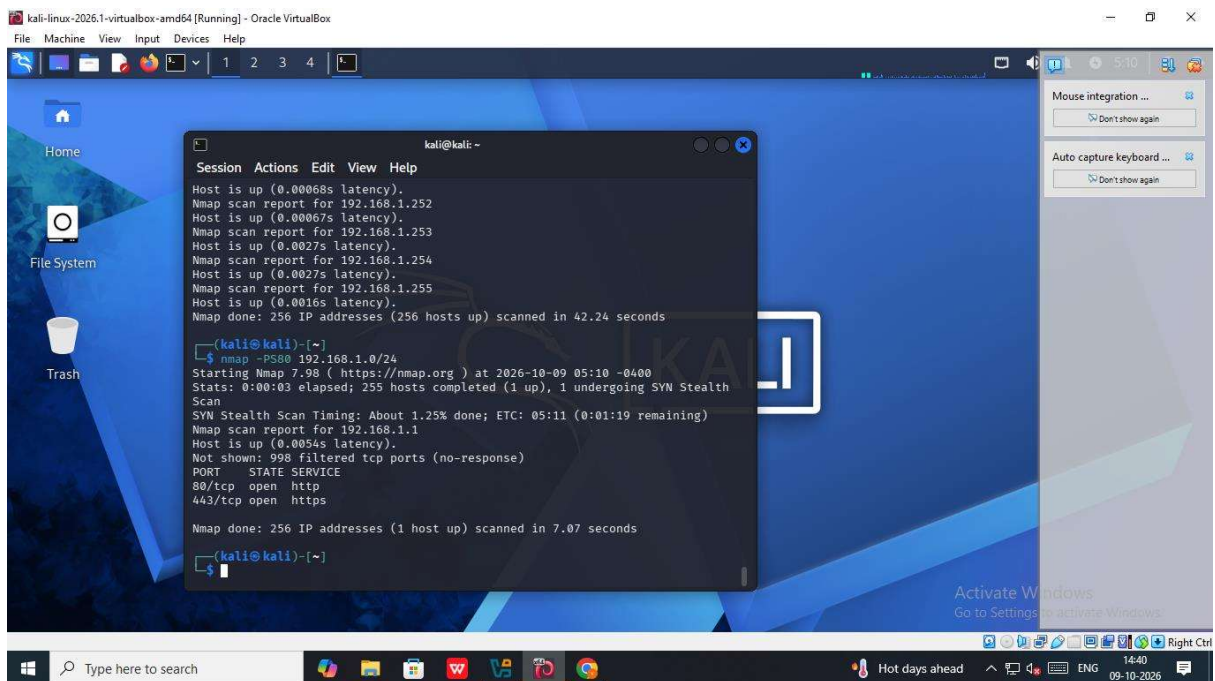


Input 2: TCP SYN Host Discovery

Command:

```
nmap -PS80 192.168.1.0/24
```

Output 2 (sample):



Input 3: UDP Host Discovery

Command:

```
sudo nmap -PU53 192.168.1.0/24
```

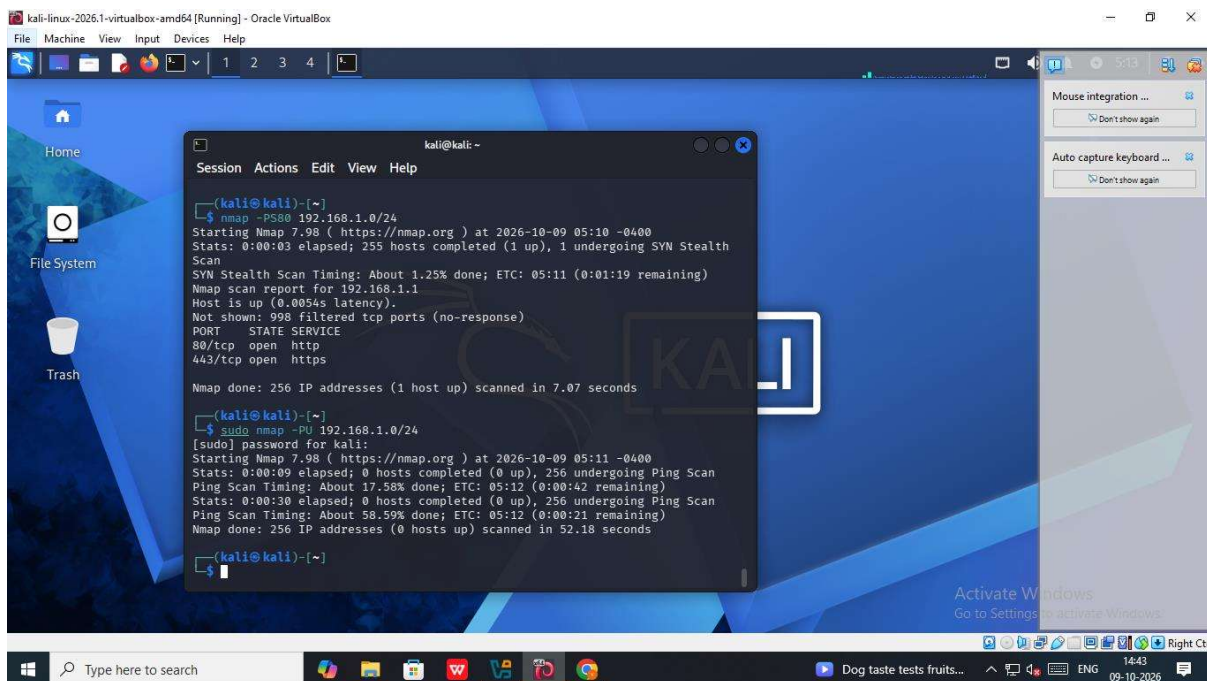
Output 3 (sample):

Input 4: ICMP Timestamp Discovery

Command:

```
nmap -PP 192.168.1.0/24
```

Output 4 (sample):



The screenshot shows a Kali Linux desktop environment with a terminal window open. The terminal displays the output of two nmap scans. The first scan is a SYN Stealth Scan (-PS80) which found one host up (192.168.1.1) with open ports 80/tcp (http) and 443/tcp (https). The second scan is a Ping Scan (-PP) which found no hosts up. The terminal window has a title bar that reads 'kali@kali -' and a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The desktop background is a blue and black abstract pattern. On the left side of the desktop, there are icons for 'Home', 'File System', and 'Trash'. On the right side, there are two utility windows: 'Mouse integration ...' and 'Auto capture keyboard ...', both with 'Don't show again' buttons. At the bottom of the screen, there is a taskbar with various application icons and a system tray showing the date and time as '14:43 09-10-2026'.

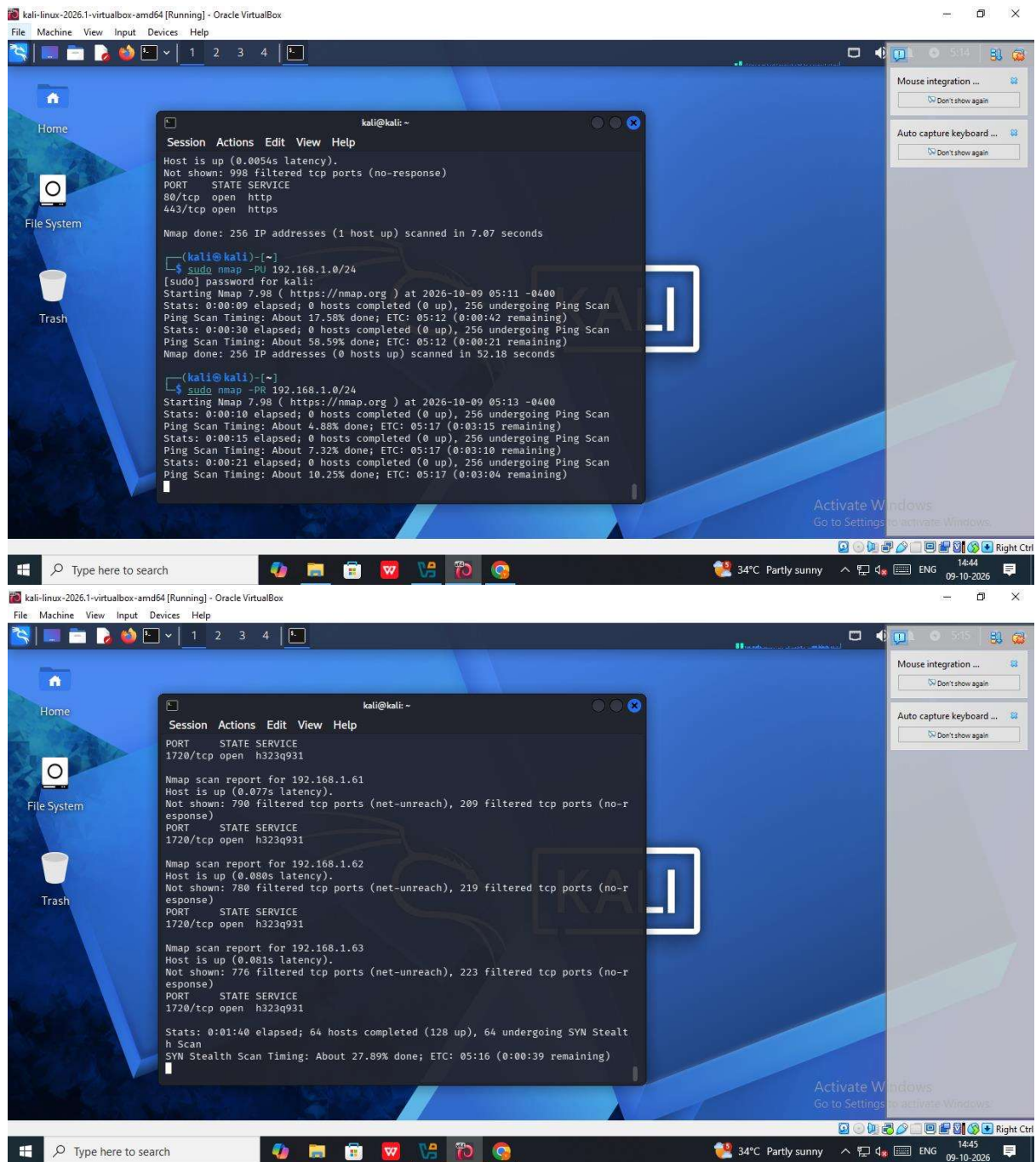
```
kali@kali ~  
$ nmap -PS80 192.168.1.0/24  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-10-09 05:10 -0400  
Stats: 0:00:03 elapsed; 255 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 1.25% done; ETC: 05:11 (0:01:19 remaining)  
Nmap scan report for 192.168.1.1  
Host is up (0.0054s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp    open  https  
  
Nmap done: 256 IP addresses (1 host up) scanned in 7.07 seconds  
  
kali@kali ~  
$ sudo nmap -PP 192.168.1.0/24  
[sudo] password for kali:  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-10-09 05:11 -0400  
Stats: 0:00:09 elapsed; 0 hosts completed (0 up), 256 undergoing Ping Scan  
Ping Scan Timing: About 17.58% done; ETC: 05:12 (0:00:42 remaining)  
Stats: 0:00:30 elapsed; 0 hosts completed (0 up), 256 undergoing Ping Scan  
Ping Scan Timing: About 58.59% done; ETC: 05:12 (0:00:21 remaining)  
Nmap done: 256 IP addresses (0 hosts up) scanned in 52.18 seconds  
  
kali@kali ~  
$
```

Input 5: ARP Host Discovery

Command:

```
sudo nmap -PR 192.168.1.0/24
```

Output 5 (sample):

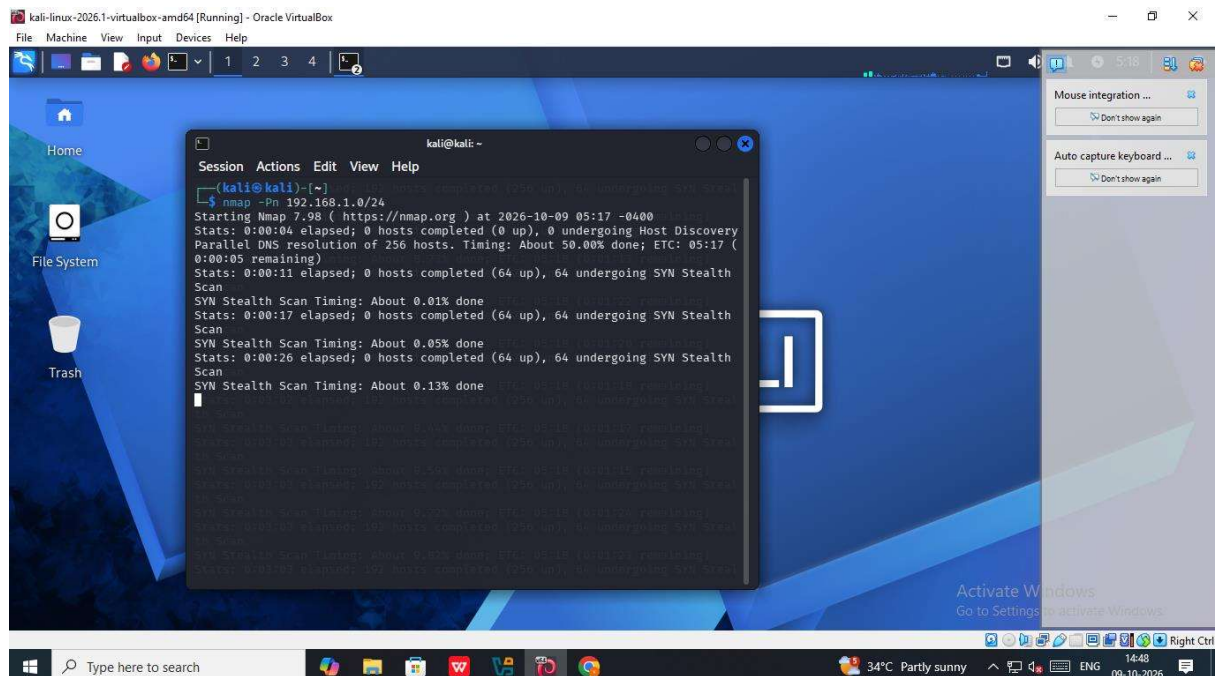


Input 6: Disable Host Discovery

Command:

```
nmap -Pn 192.168.1.0/24
```

Output 6 (sample):



Result

Nmap host discovery commands were studied to identify active hosts using ping, TCP SYN, UDP, ICMP timestamp, and ARP probes. The `-Pn` option was also demonstrated to skip host discovery.